IP

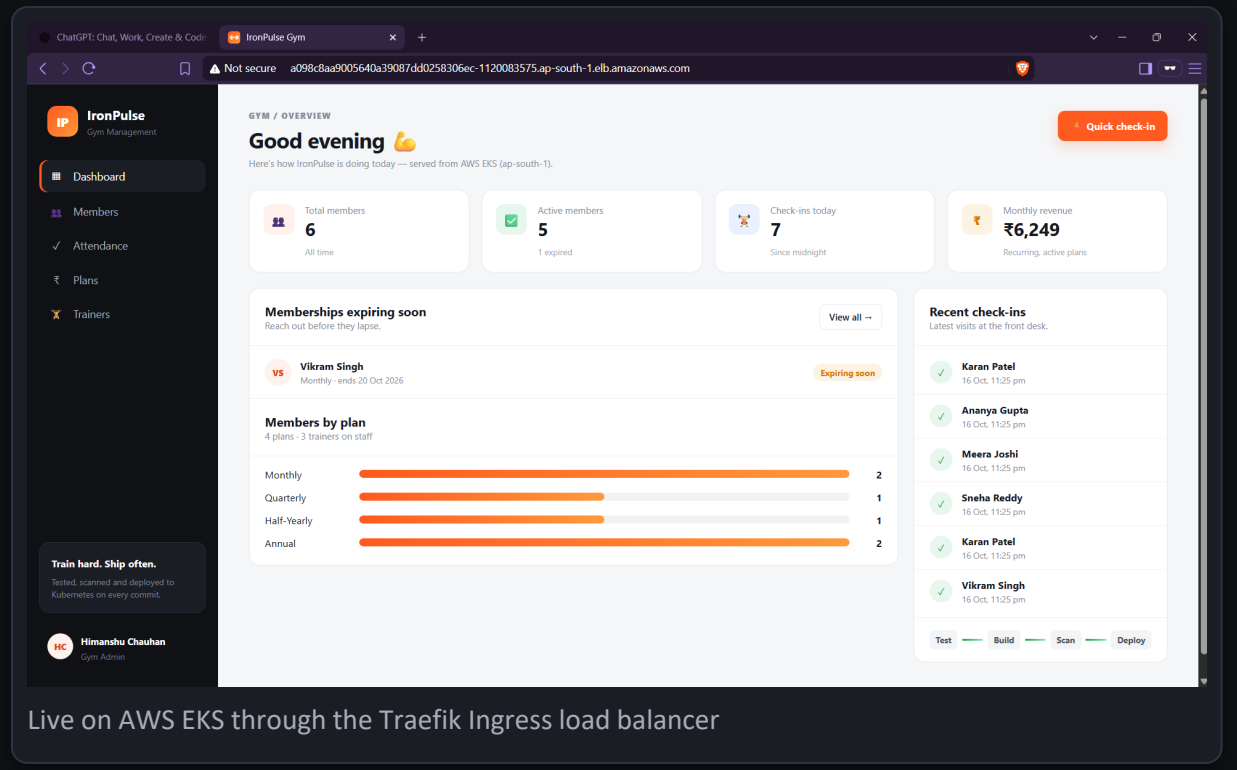
IronPulse

A gym management platform, shipped through a complete DevOps pipeline to AWS EKS

DevOps Final Capstone

Himanshu Chauhan · October 2026

github.com/Himanshu10071/devops-project



Live on AWS EKS through the Traefik Ingress load balancer

What IronPulse does

M1 · App

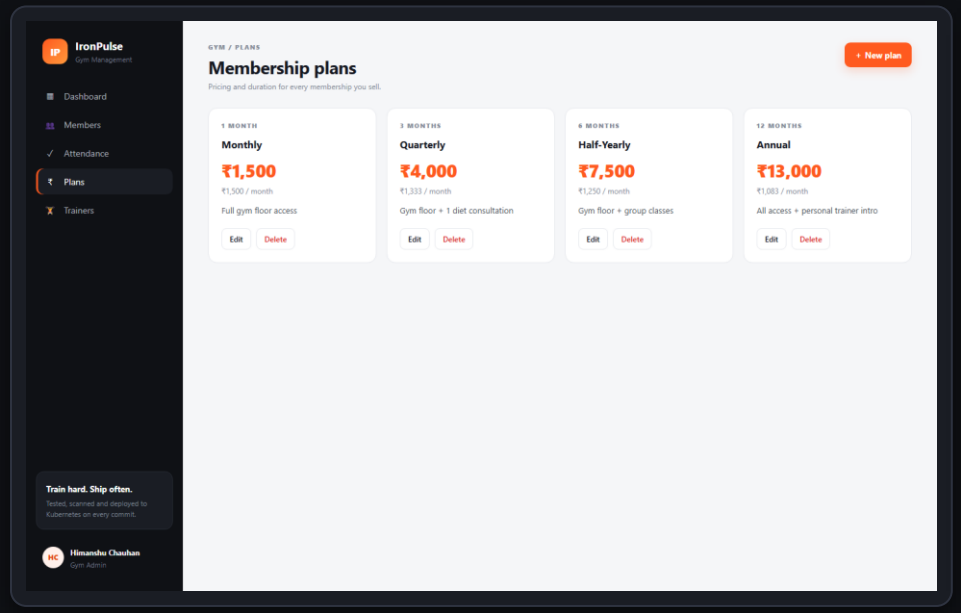
1. **Members**
   Register, search, renew, freeze; expiry is computed from the plan

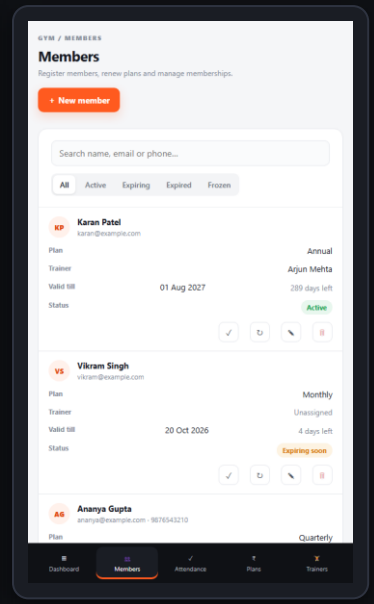
2. **Plans**
   Monthly to annual pricing; a plan in use can't be deleted

3. **Trainers**
   Coaching staff with live member counts

4. **Attendance**
   One-click check-in; expired or frozen members are refused

5. **Dashboard**
   KPIs, expiring memberships, revenue, plan mix

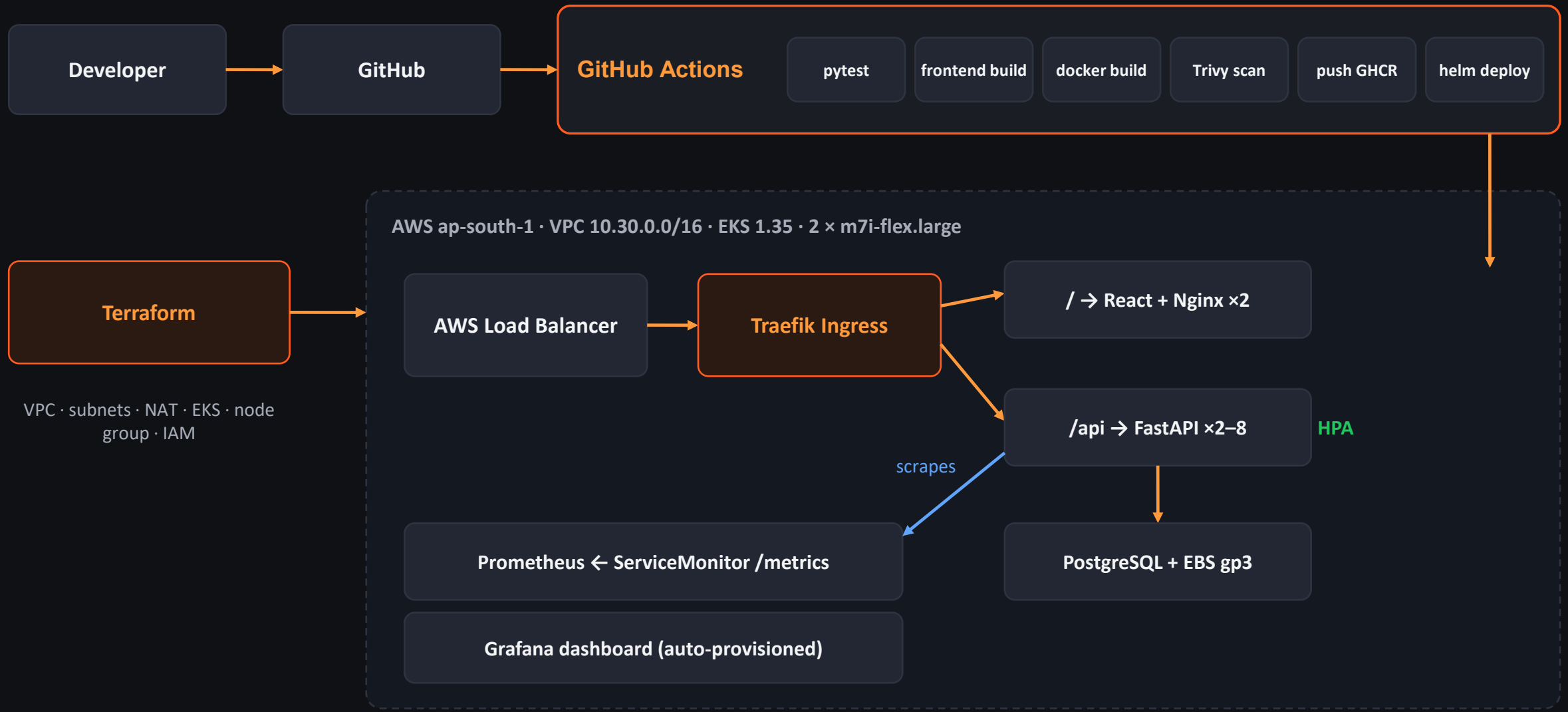




React 19 frontend · FastAPI backend · PostgreSQL 16 · 20+ REST endpoints · responsive down to phone width

From laptop to a monitored EKS cluster

Architecture



Tech stack

Overview

Frontend

React 19, Vite 8, responsive CSS

Backend

FastAPI, SQLAlchemy 2, Pydantic 2

Database

PostgreSQL 16, Alembic migrations

Testing

Pytest, 27 tests, isolated SQLite

Containers

Multi-stage Docker, non-root, Compose

CI/CD

GitHub Actions → GHCR → Helm

Security

Trivy (SHA-pinned), restricted PSA

IaC

Terraform: VPC, EKS, IAM, EBS CSI

Kubernetes

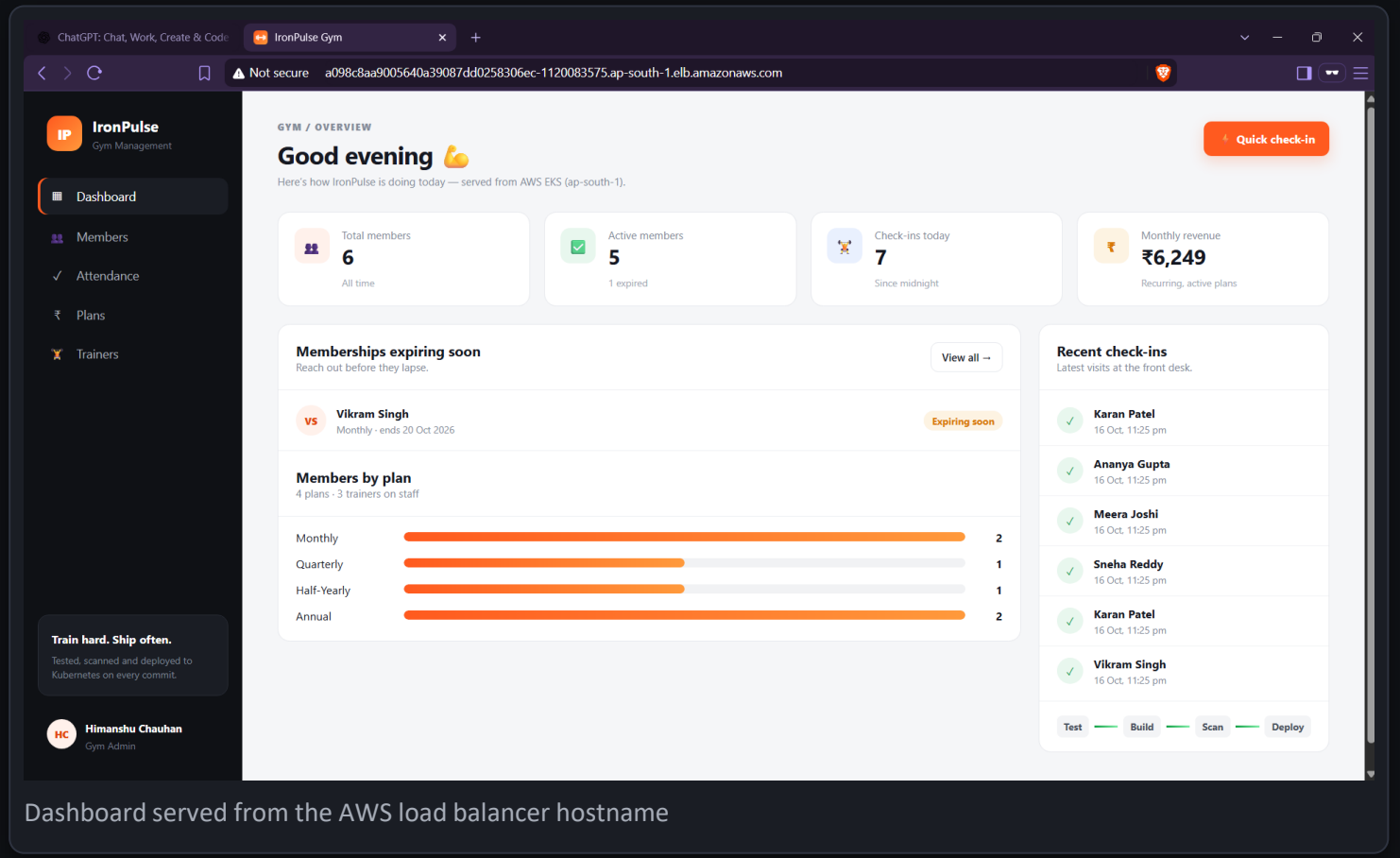
Helm, Traefik Ingress, HPA

Observability

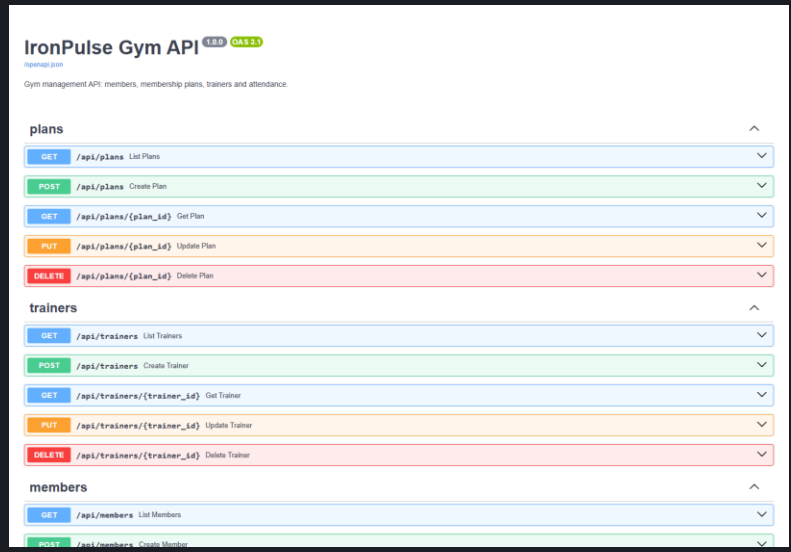
Prometheus Operator, Grafana

Live on AWS through the Ingress

M1 · M8



Dashboard served from the AWS load balancer hostname



FastAPI Swagger at /docs

- **/** → React frontend
- **/api, /docs** → FastAPI backend
- **/health** liveness, **/ready** readiness

Quality gates before any image exists

M2 · M4

27 / 27

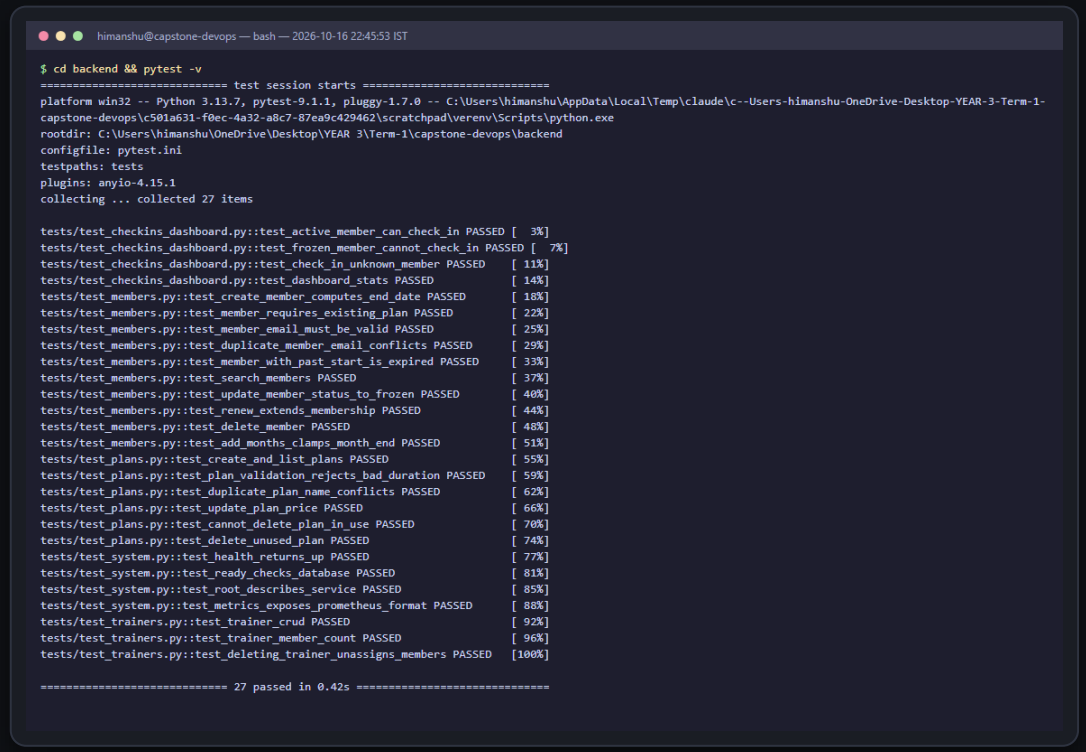
pytest tests passing, on an isolated in-memory DB

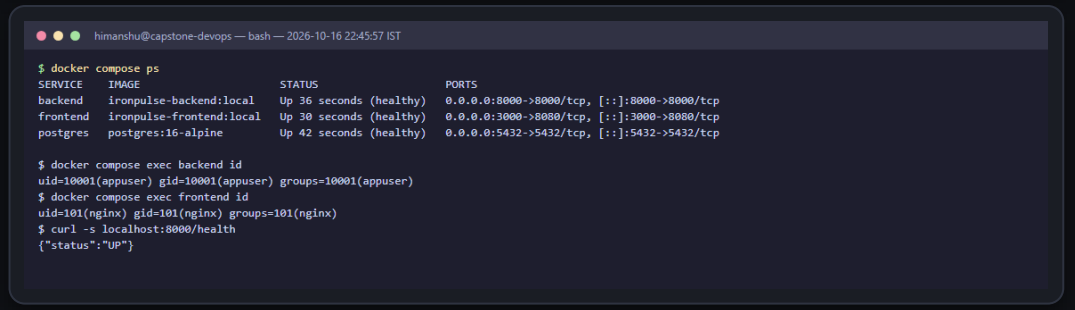
3

services from one docker compose up --build

non-root

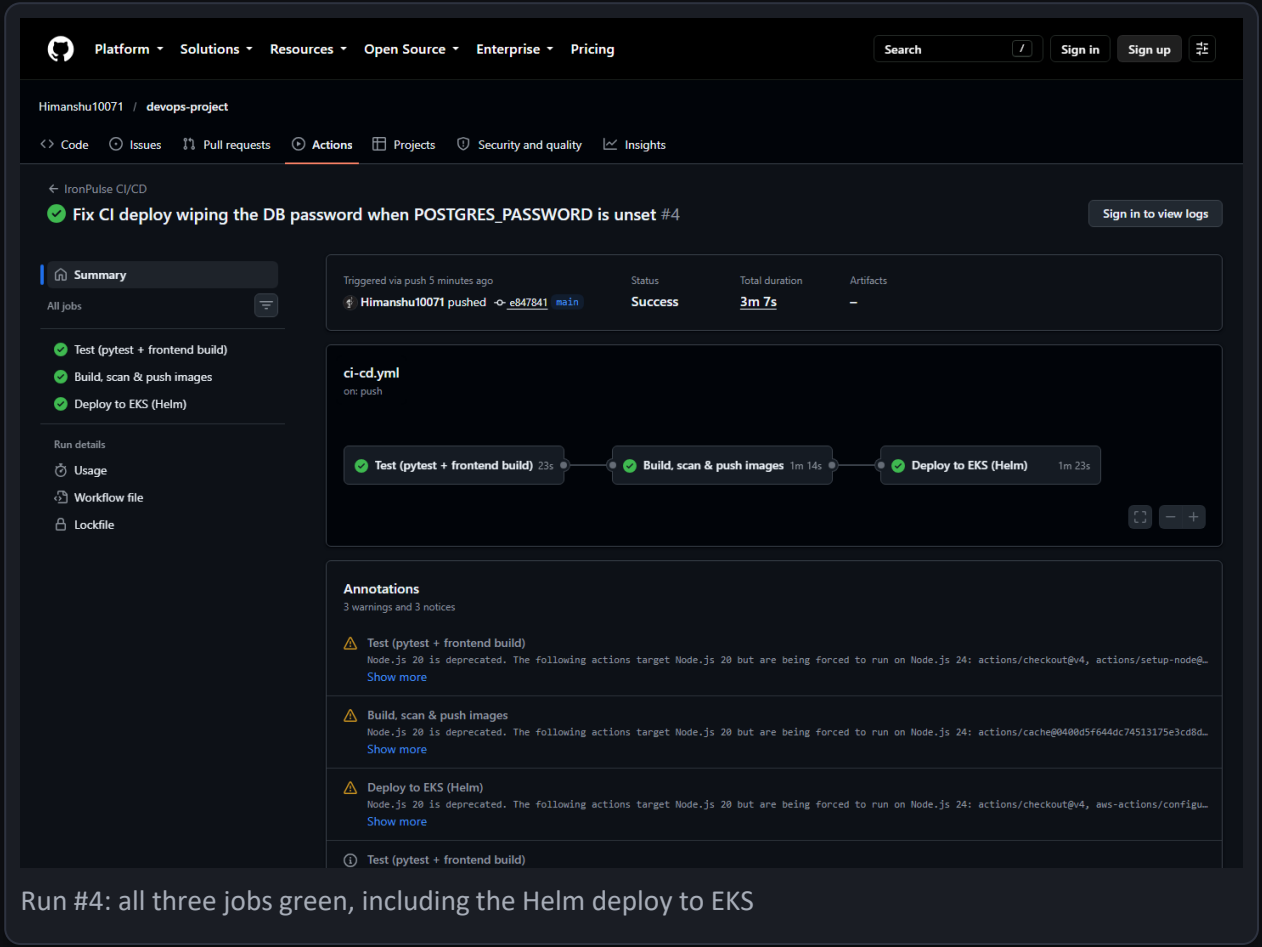
backend uid 10001 · Nginx uid 101



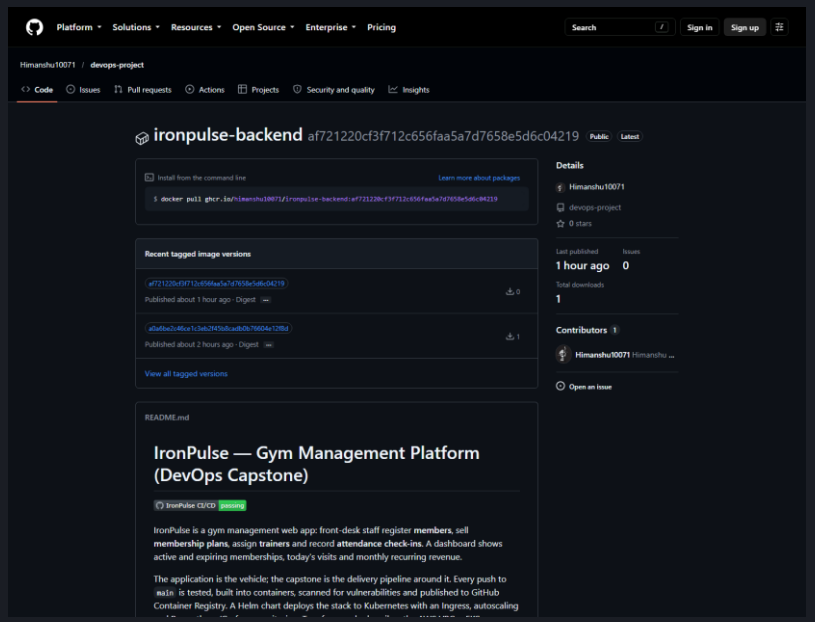


One push: test, build, scan, ship, deploy

M5 · CI/CD



Run #4: all three jobs green, including the Helm deploy to EKS



Public GHCR package, tags = commit SHA

- Tests gate the build: a failure means nothing is pushed
- Every image is tagged with the Git SHA, never latest
- Helm deploys that exact SHA and waits for readiness

Trivy: 4 HIGH CVEs found and fixed

M6 · Trivy

himanshu@capstone-devops — bash — 2026-10-16 22:47:41 IST

```
$ GitHub Actions › Build, scan & push images › Trivy scan — backend (run #2, commit af72122)
Run aquasecurity/trivy-action@57a97c7e7821a5776cebc9bb87c984fa69cba8f1
with:
  image-ref: ghcr.io/himanshu10071/ironpulse-backend:af721220cf3f712c656faa5a7d7658e5d6c04219
  severity: HIGH,CRITICAL
  ignore-unfixed: true
  exit-code: 1
  format: table
  scan-type: image
  ...
Running Trivy with options: trivy image ghcr.io/himanshu10071/ironpulse-backend:af721220cf3f712c656faa5a7d7658e5d6c04219
2026-10-16T15:47:43Z    INFO    [vuln] Vulnerability scanning is enabled
2026-10-16T15:47:43Z    INFO    [secret] Secret scanning is enabled
2026-10-16T15:47:51Z    INFO    Detected OS     family="debian" version="13.7"
2026-10-16T15:47:51Z    INFO    [debian] Detecting vulnerabilities...   os_version="13" pkg_num=87
2026-10-16T15:47:51Z    INFO    Number of language-specific files       num=1
2026-10-16T15:47:51Z    INFO    [python-pkg] Detecting vulnerabilities...
2026-10-16T15:47:51Z    WARN    Using severities from other vendors for some vulnerabilities. Read https://trivy.dev/docs/v0.69/guide/scanner/vulnerability#severity-selec
tion for details.

Report Summary

┌──────────────────────────────────────────────────────────────────────────────────────┬────────────┬─────────────────┬─────────┐
│                                        Target                                        │    Type    │ Vulnerabilities │ Secrets │
├──────────────────────────────────────────────────────────────────────────────────────┼────────────┼─────────────────┼─────────┤
│ ghcr.io/himanshu10071/ironpulse-backend:af721220cf3f712c656faa5a7d7658e5d6c04219     │   debian   │        0        │    -    │
│ (debian 13.7)                                                                        │            │                 │         │
├──────────────────────────────────────────────────────────────────────────────────────┼────────────┼─────────────────┼─────────┤
│ opt/venv/lib/python3.12/site-packages/alembic-1.20.0.dist-info/METADATA              │ python-pkg │        0        │    -    │
├──────────────────────────────────────────────────────────────────────────────────────┼────────────┼─────────────────┼─────────┤
│ opt/venv/lib/python3.12/site-packages/annotated_doc-0.0.5.dist-info/METADATA         │ python-pkg │        0        │    -    │
├──────────────────────────────────────────────────────────────────────────────────────┼────────────┼─────────────────┼─────────┤
│ opt/venv/lib/python3.12/site-packages/annotated_types-0.8.0.dist-info/METADATA       │ python-pkg │        0        │    -    │
├──────────────────────────────────────────────────────────────────────────────────────┼────────────┼─────────────────┼─────────┤
│ opt/venv/lib/python3.12/site-packages/anyio-4.15.1.dist-info/METADATA                │ python-pkg │        0        │    -    │
├──────────────────────────────────────────────────────────────────────────────────────┼────────────┼─────────────────┼─────────┤
│ opt/venv/lib/python3.12/site-packages/click-8.5.0.dist-info/METADATA                 │ python-pkg │        0        │    -    │
```

CI scan of the backend image: 0 fixable HIGH/CRITICAL

4 → 0

HIGH vulnerabilities

0

in the frontend image

- **Cause:** urllib3, msgpack, setuptools vendored inside pip
- **Fix:** drop pip from the runtime image; it's build-only
- **Supply chain:** Trivy action pinned to a commit SHA after the 2026 tag hijack
- **Runtime:** non-root pods, restricted Pod Security, least-privilege deploy user

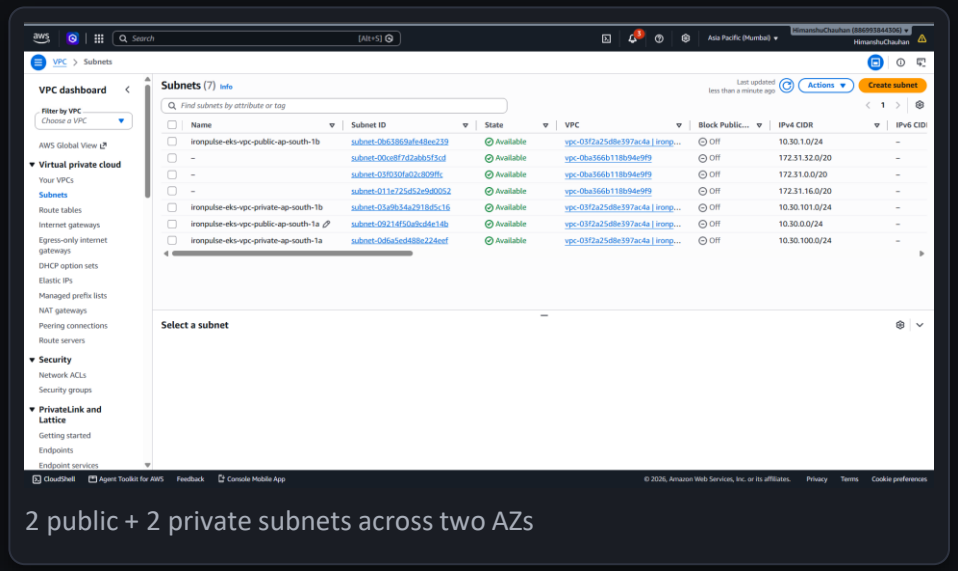
Terraform built the whole AWS stack

M7 · Terraform

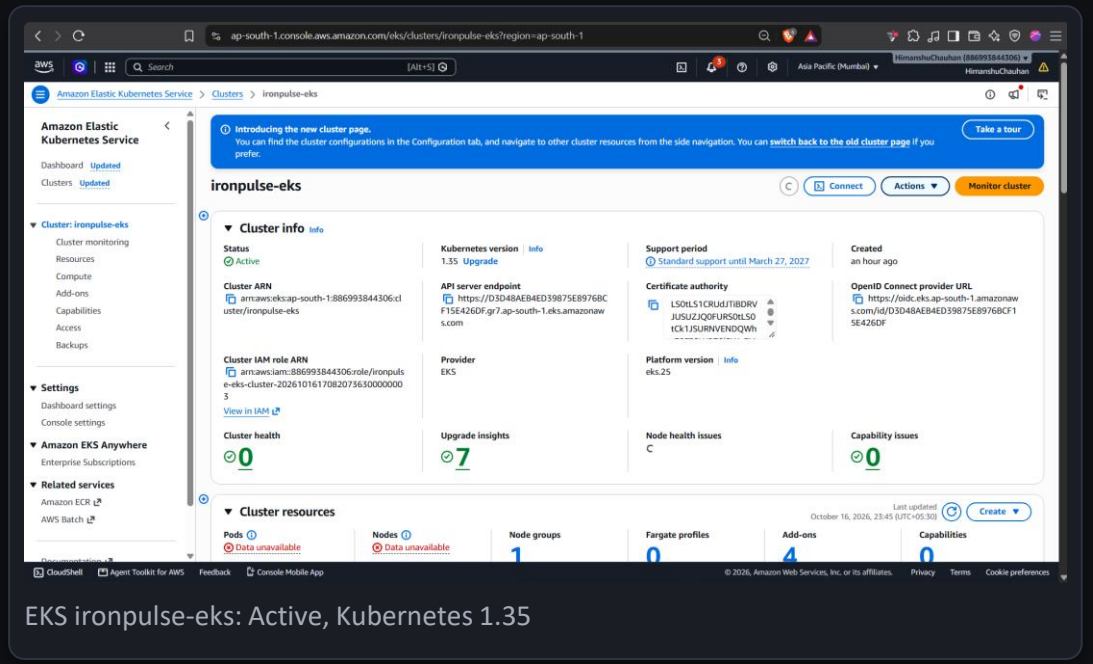
```
module.vpc.aws_route_table_association.public[0] will be created
module.vpc.aws_route_table_association.public[1] will be created
module.vpc.aws_subnet.private[0] will be created
module.vpc.aws_subnet.private[1] will be created
module.vpc.aws_subnet.public[0] will be created
module.vpc.aws_subnet.public[1] will be created
module.vpc.aws_vpc.this[0] will be created
module.eks.module.eks_managed_node_group["main"].aws_eks_node_group.this[0] will be created
module.eks.module.eks_managed_node_group["main"].aws_iam_role.this[0] will be created
module.eks.module.eks_managed_node_group["main"].aws_iam_role_policy_attachment.this["AmazonEC2ContainerRegistryReadOnly"] will be created

Plan: 64 to add, 0 to change, 0 to destroy.
Changes to Outputs:
  + ci_deployer_access_key_id     = (known after apply)
  + ci_deployer_secret_access_key = (sensitive value)
  + cluster_endpoint              = (known after apply)
  + cluster_name                  = "ironpulse-eks"
  + cluster_version               = "1.35"
  + configure_kubectl             = "aws eks update-kubeconfig --region ap-south-1 --name ironpulse-eks"
```

terraform plan: 64 resources to add



2 public + 2 private subnets across two AZs



EKS ironpulse-eks: Active, Kubernetes 1.35

- VPC 10.30.0.0/16, NAT gateway, private worker nodes
- Managed node group: 2 × m7i-flex.large (AL2023)
- EBS CSI driver via IRSA, CI deployer IAM user

Kubernetes + Helm on EKS

M8 · K8s

himanshu@capstone-devops — bash — 2026-10-16 23:44:07 IST

```
$ kubectl get nodes,pods  (EKS ironpulse-eks, ap-south-1)
$ kubectl get nodes -L node.kubernetes.io/instance-type,topology.kubernetes.io/zone
NAME                                          STATUS   ROLES    AGE   VERSION              INSTANCE-TYPE    ZONE
ip-10-30-100-194.ap-south-1.compute.internal   Ready    <none>   38m   v1.35.8-eks-b9510e3   m7i-flex.large   ap-south-1a
ip-10-30-101-115.ap-south-1.compute.internal   Ready    <none>   38m   v1.35.8-eks-b9510e3   m7i-flex.large   ap-south-1b

$ kubectl get pods -n ironpulse -o custom-columns=...
NAME                                 READY   STATUS    IMAGE                                                                                                                NODE
ironpulse-backend-8dcb7d77c-g45tp    true    Running   ghcr.io/himanshu10071/ironpulse-backend:e8478412292a5582baced3cf1ece3a30d7f10728    ip-10-30-100-194.ap-south-1.compute.internal
ironpulse-backend-8dcb7d77c-m786x    true    Running   ghcr.io/himanshu10071/ironpulse-backend:e8478412292a5582baced3cf1ece3a30d7f10728    ip-10-30-101-115.ap-south-1.compute.internal
ironpulse-frontend-565f7fbc5d-74dwk  true    Running   ghcr.io/himanshu10071/ironpulse-frontend:e8478412292a5582baced3cf1ece3a30d7f10728   ip-10-30-100-194.ap-south-1.compute.internal
ironpulse-frontend-565f7fbc5d-shgpw  true    Running   ghcr.io/himanshu10071/ironpulse-frontend:e8478412292a5582baced3cf1ece3a30d7f10728   ip-10-30-101-115.ap-south-1.compute.internal
ironpulse-postgres-0                 true    Running   postgres:16-alpine                                                                                                   ip-10-30-100-194.ap-south-1.compute.internal
```

himanshu@capstone-devops — bash — 2026-10-16 23:43:24 IST

```
$ kubectl get svc,ingress,hpa -n ironpulse
$ kubectl get svc,ingress,hpa -n ironpulse
NAME                              TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)    AGE
service/ironpulse-backend    ClusterIP   172.20.42.1      <none>        8000/TCP   19m
service/ironpulse-frontend   ClusterIP   172.20.240.124   <none>        80/TCP     19m
service/ironpulse-postgres   ClusterIP   172.20.233.166   <none>        5432/TCP   19m

NAME                                CLASS     HOSTS   ADDRESS                                                                        PORTS   AGE
ingress.networking.k8s.io/ironpulse   traefik   *       a098c8aa9005640a39087dd0258306ec-1120083575.ap-south-1.elb.amazonaws.com   80      19m

NAME                                                        REFERENCE                       TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
horizontalpodautoscaler.autoscaling/ironpulse-backend    Deployment/ironpulse-backend    cpu: 2%/60%   2         8         2          19m
horizontalpodautoscaler.autoscaling/ironpulse-frontend   Deployment/ironpulse-frontend   cpu: 2%/70%   2         4         2          19m

$ kubectl get svc -n traefik traefik
NAME      TYPE           CLUSTER-IP       EXTERNAL-IP                                                                    PORT(S)                      AGE
traefik   LoadBalancer   172.20.214.204   a098c8aa9005640a39087dd0258306ec-1120083575.ap-south-1.elb.amazonaws.com   80:32123/TCP,443:31455/TCP   22m

$ kubectl get pvc -n ironpulse
NAME                              STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
data-ironpulse-postgres-0   Bound    pvc-1c6bec05-33c4-4a48-946e-6c66abf58f4b   10Gi       RWO            gp3            <unset>                 19m
```

- 2 backend + 2 frontend pods, spread over 2 AZs
- ClusterIP Services, Traefik Ingress, Postgres StatefulSet
- Restricted Pod Security: non-root, no capabilities
- Zero-downtime rolling updates; helm rollback ready

Autoscaling under real load

M8 · HPA

2 → 4

backend pods added by the HPA

99%

CPU vs. the 60% target at peak

240 s

load test: 40 workers through the AWS load balancer

himanshu@capstone-devops — bash — 2026-10-16 23:50:10 IST

```
$ # HPA autoscaling under load (EKS)
$ URL=http://a098c8aa9005640a39087dd0258306ec-1120083575.ap-south-1.elb.amazonaws.com ./scripts/load-test.sh 240 40   # running in another terminal
$ kubectl get hpa ironpulse-backend -n ironpulse -w
NAME                REFERENCE                      TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
ironpulse-backend   Deployment/ironpulse-backend   cpu: 2%/60%   2         8         2          20m
ironpulse-backend   Deployment/ironpulse-backend   cpu: 59%/60%  2         8         2          20m
ironpulse-backend   Deployment/ironpulse-backend   cpu: 66%/60%  2         8         2          21m
ironpulse-backend   Deployment/ironpulse-backend   cpu: 99%/60%  2         8         4          21m
ironpulse-backend   Deployment/ironpulse-backend   cpu: 75%/60%  2         8         4          22m
ironpulse-backend   Deployment/ironpulse-backend   cpu: 39%/60%  2         8         4          22m
ironpulse-backend   Deployment/ironpulse-backend   cpu: 38%/60%  2         8         4          22m
ironpulse-backend   Deployment/ironpulse-backend   cpu: 29%/60%  2         8         4          23m
ironpulse-backend   Deployment/ironpulse-backend   cpu: 34%/60%  2         8         4          23m
ironpulse-backend   Deployment/ironpulse-backend   cpu: 38%/60%  2         8         4          24m
ironpulse-backend   Deployment/ironpulse-backend   cpu: 48%/60%  2         8         3          24m
ironpulse-backend   Deployment/ironpulse-backend   cpu: 46%/60%  2         8         3          24m
ironpulse-backend   Deployment/ironpulse-backend   cpu: 3%/60%   2         8         3          25m
ironpulse-backend   Deployment/ironpulse-backend   cpu: 2%/60%   2         8         3          25m
ironpulse-backend   Deployment/ironpulse-backend   cpu: 2%/60%   2         8         3          26m
```

kubectl get hpa -w during scripts/load-test.sh

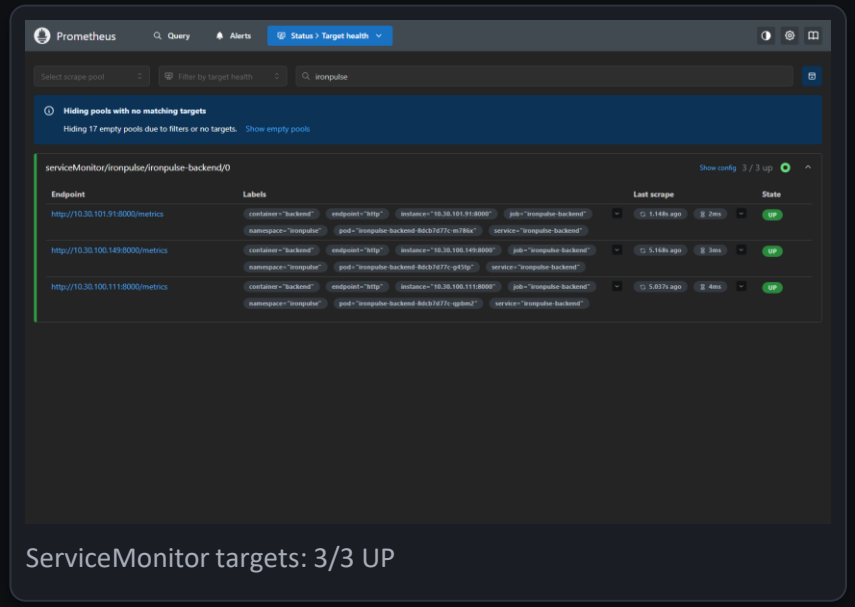
Load stops → scale-down waits for a 2-minute stabilization window, then steps back to 3 and 2 replicas.

Prometheus + Grafana watching the API

M9 · Monitoring



Auto-provisioned dashboard during the load test



ServiceMonitor targets: 3/3 UP

- /metrics: request rate, latency histograms, status codes
- Business metric: gym_checkins_total
- Alerts: backend down, >5% errors, p95 > 1 s

What broke, and how it was fixed

Lessons

No worker nodes

AWS Free plan rejected t3.medium. Switched to Free Tier eligible m7i-flex.large.

Root can't use kubectl

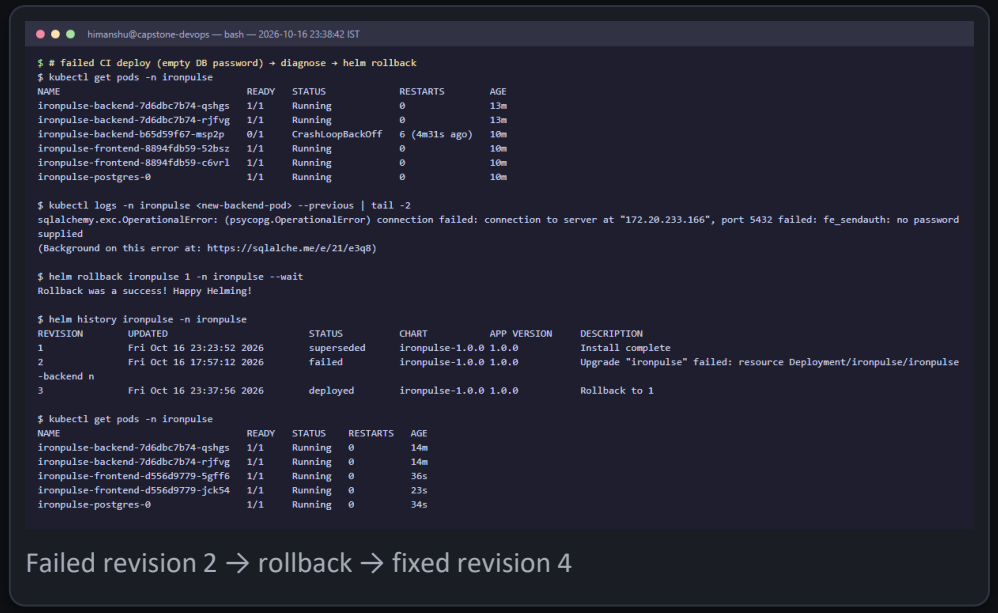
EKS rejects root as a principal. Terraform now creates a scoped deployer IAM user.

Deploy crash-looped

CI passed an empty DB password. Old pods kept serving; helm rollback, then fixed the workflow + chart.

Lab pods never started

Restricted Pod Security blocked them. Lab manifests now set a compliant securityContext.

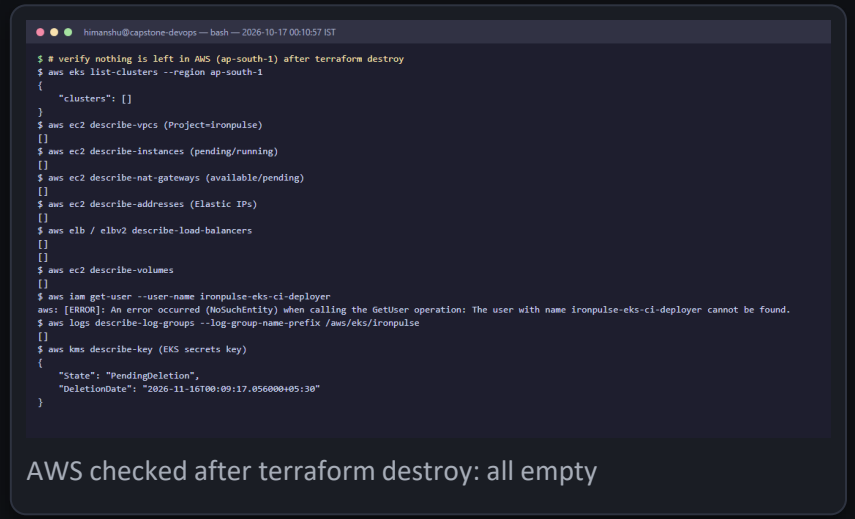


Failed revision 2 → rollback → fixed revision 4

Every incident is reproducible from the repo and documented in the README.

Every rubric module, delivered and verified

✓ M1 App	✓ M2 Tests	✓ M3 Git	✓ M4 Docker	✓ M5 CI/CD
✓ M6 Trivy	✓ M7 Terraform	✓ M8 K8s + Helm	✓ M9 Monitoring	✓ M10 Docs



AWS checked after terraform destroy: all empty

Repository github.com/Himanshu10071/devops-project

Evidence 34 screenshots in docs/screenshots · rubric map in docs/SUBMISSION.md

Torn down after verification: no clusters, nodes, NAT gateway, load balancers or volumes left running.

Thank you